

DBACT THEOREM 1

Proofs and Supporting Results

Boundary-measure-induced Local-CVT practical stability near a selected nondegenerate local branch

Branch	DBACT-research-v3
Baseline HEAD	64e1905e092d19fd9dcbf0e77fbd39f55f0562ba
Proof status	Conditional local theorem; A8 interval certificate not supplied
Wolfram	Connector exact checks passed; 9 local scripts exit 0
Regression	477 passed, 3 skipped

CLAIM BOUNDARY

No global PL. No formal caging. No finite-time phase completion. No unconditional safety or cooperative-transport success.

Contents

1	Scope and frozen claim	3
1.1	Deliberate exclusions	3
2	Notation	4
2.1	Kernel constants	4
2.2	Units	4
3	Assumptions A1-A10	5
4	Lemma 1 - boundary and map to density consistency	6
4.1	Statement	6
4.2	Proof	6
4.3	Evidence boundary	9
5	Lemma 2 - mass, quadrature and centroid perturbation	10
5.1	Statement	10
5.2	Proof	10
6	Proposition 1 - ideal Local-CVT gradient and descent	13
6.1	Statement	13
6.2	Proof	13
7	Proposition 2 - aggregate implementation disturbance	15
7.1	Statement	15
7.2	Proof	15
8	Proposition 3 - reference rate and local regularity	17
8.1	Target and density rate	17
8.2	Cost and optimal-value rate	17
8.3	Local PL and quadratic growth (M5)	18
9	Theorem 1 - local practical stability	19
9.1	Statement	19
9.2	Proof - every inequality named	19
9.3	Status	20
10	Corollaries	21

10.1 Frozen perfect information	21
10.2 Sampled-data corollary (M7)	21
11 Limits and evidence status	23
11.1 Evidence classes	23
11.2 Fail-closed runtime gates	23
11.3 Claim audit	23
A Constant ledger	24
B Wolfram verification manifest	25
B.1 What computer algebra did not verify	25
B.2 Reference	25

1 Scope and frozen claim

Bounded local-map disagreement and bounded reference variation imply practical stability of the boundary-induced Local-CVT allocation dynamics around a nondegenerate local CVT branch.

The agents are first-order, $\dot{p}_i = u_i$, and the nominal ENCLOSE law is $u_i^{\text{nom}} = -k_c(p_i - \hat{c}_i)$. The ideal objective is the *saturated* limited-range coverage cost

$$F_R(r) = \min\{r^2, R_\ell^2\}, \quad H^*(P, t) = \int_D \min_i F_R(\|q - p_i\|) \phi^*(q, t) dq$$

Saturation is not cosmetic. With the unsaturated truncated cost $\int_{V_i^\ell} \|q - p_i\|^2 \phi^* dq$ the integration domain moves with p_i , and the Reynolds boundary term on the circle $\|q - p_i\| = R_\ell$ does not cancel; the standard CVT gradient identity is then false. Section 5 shows the cancellation that saturation buys.

1.1 Deliberate exclusions

This supplement does not prove any of the following.

- global CVT optimality, or convergence from arbitrary initial configurations;
- finite-time SEARCH-to-HOLD completion;
- geometric or topological caging;
- contact mechanics, or unconditional cooperative-transport success;
- CBF forward invariance when feasibility, speed-cap, or deviation contracts are violated;
- a uniform reduced-Hessian bound for an unspecified local branch.

Collision safety is a separate result and is deliberately not a dependency of Theorem 1.

2 Notation

$D \subset \mathbb{R}^2$ is a fixed compact rectangle of width W_D and height H_D ; $P = (p_1, \dots, p_N) \in \mathbb{R}^{2N}$ stacks the agent positions. All spatial norms are Euclidean, and the stacked norm is the Euclidean product norm

$$\|d\|^2 = \sum_{i=1}^N \|d_i\|^2$$

$\Gamma(t)$ is the true object boundary, with perimeter $P_\Gamma \leq P_{\max}$, and $\xi(s, t)$ is the offset-boundary target parameterised by arc length s . The ideal boundary measure is the pushforward of arc length, and the ideal density is its Gaussian convolution above a positive floor:

$$\nu_t^* = \xi(\cdot, t)_\# ds, \quad \phi^*(q, t) = \phi_0 + (K_\sigma * \nu_t^*)(q) = \phi_0 + \int_{\Gamma(t)} K_\sigma(q - \xi(s, t)) ds$$

The kernel is the *unnormalised* Gaussian $K_\sigma(x) = \exp(-\|x\|^2/(2\sigma^2))$. The local cell, its mass and its centroid are

$$V_i^\ell(P) = \text{Vor}_i(P) \cap D \cap B(p_i, R_\ell), \quad m_i^* = \int_{V_i^\ell} \phi^* dq, \quad c_i^* = \frac{1}{m_i^*} \int_{V_i^\ell} q \phi^*(q, t) dq$$

$\hat{\phi}_i$ is agent i 's mapped density, $\hat{c}_i$ its numerically computed centroid, $\mathcal{C}^*(t)$ the selected local minimiser branch inside a fixed compact regular tube X , and

$$V(P, t) = H^*(P, t) - H_{\min}^*(t) \geq 0, \quad H_{\min}^*(t) = \min_{P \in X} H^*(P, t)$$

2.1 Kernel constants

These three constants are used throughout; all are Wolfram-verified (entries W-RR-01 to W-RR-03) and derived in Step 1 of Lemma 1.

$$\|K_\sigma\|_1 = 2\pi\sigma^2, \quad \|\nabla K_\sigma\|_1 = \sqrt{2}\pi^{3/2}\sigma, \quad \|\nabla K_\sigma\|_\infty = \frac{e^{-1/2}}{\sigma} \quad (\text{K})$$

2.2 Units

Arc length is in metres and K_σ is dimensionless, so ϕ has units of m, an L^1 density error and a cell mass have units m^3 , a shifted first moment m^4 , H^* has m^5 , ∇H^* has m^4 , and a command disturbance has m/s . The ledger in Appendix A carries the unit of every constant so that geometric length and density mass are never silently added.

3 Assumptions A1-A10

ID	Assumption	How it is discharged
A1	D is a fixed rectangle and the solution remains in a fixed compact configuration tube X on the claimed interval.	assumption / invariant-domain check
A2	Pairwise separation is at least $d_s > 0$, and $R_{\text{comm}} \geq 2R_\ell$.	runtime certificate + config validator (fail-closed)
A3	$\Gamma(t)$ is closed and piecewise C^2 with $P_\Gamma \leq P_{\text{max}}$; version 1 uses a rigid boundary with bounded twist.	assumption / scenario certificate
A4	$\phi_0 > 0$, $\sigma > 0$, and the unnormalised Gaussian above is used.	analytic + config
A5	Map weights are nonnegative and a certificate supplies $\varepsilon_b, \varepsilon_n, \varepsilon_d$, voxel v , h_s , E_w , M_{drop} and s_σ as upper bounds.	runtime / offline certificate
A6	Local CVT uses a rectangular midpoint grid of side at most h , and $\eta_m < m_{\text{min}}$.	code contract + fail-closed test
A7	Gap, frontier and transport density biases are disabled in theorem mode, or their stacked command effect is included in δ .	config + perturbation ledger
A8	On a symmetry-reduced convex chart of the selected branch, $\nabla^2 H_{\text{red}}^*(z, t) \succeq m_H I$ uniformly, $m_H > 0$.	CONDITIONAL - no interval certificate supplied
A9	Voronoi ties and the threshold set $\ q - p_i\ = R_\ell$ have measure zero, and the solution stays on one regular branch tube.	local theorem assumption
A10	ZOH, numerical, tracking and safety-filter deviations from the ideal field have certified stacked-norm bounds.	analytic + runtime certificate

STATUS OF THE ASSUMPTION SET

The implication A1-A10 $\Rightarrow$ Theorem 1 is proved. A5, A8, A9 and A10 are not facts about every run: they are explicit assumptions and certificate gates. The result is therefore a complete conditional local theorem, not an unconditional system-success claim.

4 Lemma 1 - boundary and map to density consistency

4.1 Statement

Assume A3-A5, let the arc partition satisfy $0 \leq \Delta s_k \leq h_s$, and suppose every source discarded by the restriction step is at least $s_\sigma \sigma$ away from every query point in $B(p_i, R_\ell) \cap D$. Define the target, weight and tail errors

$$\varepsilon_{\xi,i} = \varepsilon_{b,i} + \varepsilon_{d,i} + 2d_{\max} \sin\left(\frac{\varepsilon_{n,i}}{2}\right), \quad \varepsilon_{x,i} = \varepsilon_{\xi,i} + \sqrt{2} v$$

$$E_{w,i} = L_{\text{miss},i} + M_{\text{spur},i} + \sum_{k \in \text{matched}} |\hat{w}_{ik} - \Delta s_k|$$

Then

$$\|\hat{\phi}_i - \phi^*\|_{L^1(B(p_i, R_\ell) \cap D)} \leq \varepsilon_{\phi,i} := L_{K,1} P_{\max} \left(\varepsilon_{\xi,i} + \sqrt{2} v + \frac{L_T h_s}{2} \right) + 2\pi \sigma^2 E_{w,i} + \varepsilon_{\text{tail},i} \quad (\text{L1})$$

with $L_{K,1} := \|\nabla K_\sigma\|_1$ and

$$\varepsilon_{\text{tail},i} \leq \pi R_\ell^2 M_{\text{drop},i} e^{-s_\sigma^2/2}$$

WHY THREE MEASURES

The comparison is made after convolution, through an exact discrete intermediate measure. It is never claimed that a continuous arc-length measure converges to a sum of Dirac masses in total variation - that statement is false.

$$\nu_t^* \longrightarrow \nu_{h,t}^* = \sum_k \Delta s_k \delta_{\xi_k^*} \longrightarrow \hat{\nu}_{i,t} = \sum_k \hat{w}_{ik} \delta_{\hat{\xi}_{ik}}$$

4.2 Proof

Step 1. The three kernel constants

The mass separates into one-dimensional Gaussians:

$$\|K_\sigma\|_1 = \int_{\mathbb{R}^2} e^{-\|x\|^2/(2\sigma^2)} dx = \left(\int_{\mathbb{R}} e^{-x^2/(2\sigma^2)} dx \right)^2 = (\sigma \sqrt{2\pi})^2 = 2\pi \sigma^2$$

Since $\nabla K_\sigma(x) = -\sigma^{-2} x K_\sigma(x)$, its magnitude is radial, $\|\nabla K_\sigma(x)\| = (r/\sigma^2) e^{-r^2/(2\sigma^2)}$ with $r = \|x\|$. Integrating in polar coordinates,

$$\|\nabla K_\sigma\|_1 = \int_0^\infty \frac{r}{\sigma^2} e^{-r^2/(2\sigma^2)} 2\pi r dr = \frac{2\pi}{\sigma^2} \int_0^\infty r^2 e^{-r^2/(2\sigma^2)} dr = \frac{2\pi}{\sigma^2} \cdot \sigma^3 \sqrt{\frac{\pi}{2}} = \sqrt{2} \pi^{3/2} \sigma$$

For the sup-norm, write $g(r) = (r/\sigma^2)e^{-r^2/(2\sigma^2)}$. Then

$$g'(r) = \frac{1}{\sigma^2}e^{-r^2/(2\sigma^2)}\left(1 - \frac{r^2}{\sigma^2}\right) = 0 \iff r = \sigma \quad (r > 0)$$

and $g(0) = 0$, $g(r) \rightarrow 0$ as $r \rightarrow \infty$, so the interior stationary point is the maximum:

$$\|\nabla K_\sigma\|_\infty = g(\sigma) = \frac{\sigma}{\sigma^2}e^{-1/2} = \frac{e^{-1/2}}{\sigma}$$

Step 2. Translation inequality in L^1

This is the workhorse of the whole lemma: moving a source by a distance costs at most $\|\nabla K_\sigma\|_1$ times that distance, in L^1 . For $a, b \in \mathbb{R}^2$ the fundamental theorem of calculus along the segment $\tau \mapsto b + \tau(a - b)$ gives

$$K_\sigma(x - a) - K_\sigma(x - b) = - \int_0^1 \nabla K_\sigma(x - b - \tau(a - b)) \cdot (a - b) d\tau$$

Taking absolute values, integrating in x and exchanging the order of integration by Tonelli (the integrand is nonnegative),

$$\|K_\sigma(\cdot - a) - K_\sigma(\cdot - b)\|_1 \leq \|a - b\| \int_0^1 \int_{\mathbb{R}^2} \|\nabla K_\sigma(x - b - \tau(a - b))\| dx d\tau = \|\nabla K_\sigma\|_1 \|a - b\| \quad (\text{T})$$

the inner integral being translation-invariant and therefore equal to $\|\nabla K_\sigma\|_1$ for every τ .

Step 3. Target error

With $\xi = b + d n$ and the observed $\widehat{\xi} = \widehat{b} + \widehat{d} \widehat{n}$, the triangle inequality gives

$$\|\widehat{\xi} - \xi\| \leq \|\widehat{b} - b\| + |\widehat{d} - d| + d_{\max} \|\widehat{n} - n\|$$

Both normals are unit vectors, so if they subtend an angle θ then

$$\|\widehat{n} - n\| = \sqrt{2 - 2 \cos \theta} = 2 \sin \frac{\theta}{2}, \quad 0 \leq \theta \leq \pi$$

which is monotone in θ ; substituting $\theta \leq \varepsilon_n$ yields ε_ξ . Replacing a target by a representative of the same square voxel of side v moves it by at most the voxel diagonal $\sqrt{2}v$, which gives $\varepsilon_x = \varepsilon_\xi + \sqrt{2}v$.

SCOPE OF THE VOXEL TERM

$\sqrt{2}v$ is valid only while the fused representative stays inside its own voxel. If the fusion law can move it outside, the bound must be re-derived from that law; it must not be reused as written.

Step 4. Curve quadrature

Compare the exact discrete measure with the continuous one. On arc k , of length Δs_k and with representative ξ_k^* ,

$$K_\sigma * \nu_h^* - K_\sigma * \nu^* = \sum_k \int_{\text{arc}_k} \left[K_\sigma(\cdot - \xi_k^*) - K_\sigma(\cdot - \xi(s)) \right] ds$$

Apply (T) inside the sum. If ξ is L_T -Lipschitz in arc length then $\|\xi_k^* - \xi(s)\| \leq L_T |s - s_k|$, and for any representative in the arc $\int_{\text{arc}_k} |s - s_k| ds \leq \Delta s_k^2/2$. Hence

$$\|K_\sigma * \nu_h^* - K_\sigma * \nu^*\|_1 \leq L_{K,1} L_T \sum_k \frac{\Delta s_k^2}{2}$$

Because $0 \leq \Delta s_k \leq h_s$ we have $\Delta s_k^2 \leq h_s \Delta s_k$, so the sum telescopes into the perimeter (Wolfram entry W-RR-05):

$$\sum_k \Delta s_k^2 \leq h_s \sum_k \Delta s_k = h_s P_\Gamma \leq h_s P_{\max} \implies \|K_\sigma * \nu_h^* - K_\sigma * \nu^*\|_1 \leq \frac{1}{2} L_{K,1} L_T P_{\max} h_s$$

Choosing the arc midpoint would give the sharper factor $\Delta s_k^2/4$; the certificate uses the conservative $1/2$ so that it stays valid for any registered representative inside the arc.

Step 5. Map displacement and weight mismatch

Match the observed atoms to the exact ones and split the difference into a displacement part, a matched-weight part, and missing/spurious mass:

$$\hat{\nu}_i - \nu_h^* = \underbrace{\sum_{\text{matched}} \Delta s_k (\delta_{\hat{\xi}_k} - \delta_{\xi_k^*})}_{\text{displacement}} + \underbrace{\sum_{\text{matched}} (\hat{w}_k - \Delta s_k) \delta_{\hat{\xi}_k}}_{\text{weight}} - \sum_{\text{missed}} \Delta s_k \delta_{\xi_k^*} + \sum_{\text{spurious}} \hat{w}_k \delta_{\hat{\xi}_k}$$

Convolve and bound each group. By (T), the displacement group costs

$$\left\| \sum_{\text{matched}} \Delta s_k (K_\sigma(\cdot - \hat{\xi}_k) - K_\sigma(\cdot - \xi_k^*)) \right\|_1 \leq L_{K,1} \varepsilon_x \sum_k \Delta s_k \leq L_{K,1} P_{\max} \varepsilon_x$$

The remaining three groups are bounded by their total variation times the kernel mass, because $\|w K_\sigma(\cdot - y)\|_1 = |w| \|K_\sigma\|_1$:

$$\|K_\sigma * (\text{weight} + \text{missed} + \text{spurious})\|_1 \leq \|K_\sigma\|_1 \left(L_{\text{miss}} + M_{\text{spur}} + \sum_{\text{matched}} |\hat{w}_k - \Delta s_k| \right) = 2\pi\sigma^2 E_w$$

Step 6. Restriction tail

The restriction step discards far sources. It does not do so for free: the Gaussian has no compact support. A discarded atom of weight $\hat{w}_k$ lies at distance at least $s_\sigma \sigma$ from every query q in the disk, so pointwise

$$K_\sigma(q - \hat{\xi}_k) = \exp\left(-\frac{\|q - \hat{\xi}_k\|^2}{2\sigma^2}\right) \leq \exp\left(-\frac{s_\sigma^2}{2}\right)$$

Summing the discarded weights and integrating over an area no larger than πR_ℓ^2 ,

$$\varepsilon_{\text{tail},i} = \int_{B(p_i, R_\ell) \cap D} \sum_{\text{dropped}} \hat{w}_k K_\sigma(q - \hat{\xi}_k) dq \leq \pi R_\ell^2 M_{\text{drop},i} e^{-s_\sigma^2/2}$$

Step 7. Assembly

The common base density ϕ_0 appears in both $\hat{\phi}_i$ and ϕ^* and cancels. Adding Steps 4, 5 and 6 by the triangle inequality and restricting the global L^1 estimate to $B(p_i, R_\ell) \cap D$,

$$\begin{aligned} \|\hat{\phi}_i - \phi^*\|_{L^1} &\leq \|K_\sigma * \hat{\nu}_i - K_\sigma * \nu_h^*\|_1 + \|K_\sigma * \nu_h^* - K_\sigma * \nu^*\|_1 + \varepsilon_{\text{tail},i} && \text{triangle inequality} \\ &\leq L_{K,1} P_{\max} \varepsilon_{x,i} + 2\pi\sigma^2 E_{w,i} + \frac{1}{2} L_{K,1} L_T P_{\max} h_s + \varepsilon_{\text{tail},i} && \text{Steps 4-6} \\ &= L_{K,1} P_{\max} \left(\varepsilon_{\xi,i} + \sqrt{2} v + \frac{L_T h_s}{2} \right) + 2\pi\sigma^2 E_{w,i} + \varepsilon_{\text{tail},i} = \varepsilon_{\phi,i} && \text{definition of } \varepsilon_{x,i} \end{aligned}$$

which is (L1). ■

4.3 Evidence boundary

Wolfram checked the two kernel integrals exactly, and the sup-norm, the normal-angle identity and the bounded-step inequality symbolically. The pushforward construction, the translation inequality (T), the curve quadrature, the atom matching of Step 5 and the tail decomposition are the analytic argument above; no computer-algebra output is offered in their place.

5 Lemma 2 - mass, quadrature and centroid perturbation

5.1 Statement

Assume A1-A6 and the exact-cell contract $R_{\text{comm}} \geq 2R_\ell$. Put

$$r_0 = \min\left\{R_\ell, \frac{d_s}{2}, \frac{W_D}{2}, \frac{H_D}{2}\right\}, \quad m_{\min} = \frac{\phi_0 \pi r_0^2}{4} > 0, \quad m_{\max} = \pi R_\ell^2 (\phi_0 + P_{\max})$$

$$\hat{\phi}_{\max} = \phi_0 + P_{\max} + E_{w,i}, \quad \hat{L}_\phi \leq (P_{\max} + E_{w,i}) \frac{e^{-1/2}}{\sigma}$$

For a midpoint mesh of side at most h set $r_h = h/\sqrt{2}$ (the half-diagonal), $A_V \leq \pi R_\ell^2$ and $P_V \leq 2\pi R_\ell$, and define

$$\eta_m = \hat{L}_\phi r_h A_V + \hat{\phi}_{\max} (2P_V r_h + \pi r_h^2)$$

$$\eta_a = (\hat{\phi}_{\max} + R_\ell \hat{L}_\phi) r_h A_V + R_\ell \hat{\phi}_{\max} (2P_V r_h + \pi r_h^2)$$

If the gate $\eta_m < m_{\min}$ holds, set

$$\varepsilon_q = \frac{\eta_a + R_\ell \eta_m}{m_{\min} - \eta_m}$$

Then the implementation centroid obeys

$$\|\hat{c}_i - c_i^*\| \leq \varepsilon_{c,i} := \frac{2R_\ell \varepsilon_{\phi,i}}{m_{\min}} + \varepsilon_{q,i} + \varepsilon_{G,i} \quad (\text{L2})$$

In strict synchronised theorem mode $\varepsilon_{G,i} = 0$. Delayed or missing neighbour positions require a separate geometric cell-error certificate; that error must not be hidden inside $\varepsilon_{\phi,i}$.

5.2 Proof

Step 1. Mass floor

First, separation alone puts a whole disk inside agent i 's Voronoi cell. Let $\|q - p_i\| < d_s/2$. For any $j \neq i$,

$$\|q - p_j\| \geq \|p_i - p_j\| - \|q - p_i\| \geq d_s - \frac{d_s}{2} = \frac{d_s}{2} > \|q - p_i\|$$

so q is strictly closer to p_i , i.e. $B(p_i, d_s/2) \subset \text{Vor}_i(P)$. Intersecting with the local disk and the rectangle, $V_i^\ell \supseteq B(p_i, r_0) \cap D$. The worst placement of p_i in a rectangle is a corner, which retains a quarter disk, so $|V_i^\ell| \geq \pi r_0^2/4$. Since $\phi^* \geq \phi_0$,

$$m_i^* = \int_{V_i^\ell} \phi^* dq \geq \phi_0 \cdot \frac{\pi r_0^2}{4} = m_{\min} > 0$$

NON-RECTANGULAR WORKSPACES

The quarter-disk constant is specific to a rectangle. For another workspace, replace it with a uniform interior-cone constant; the quarter-disk value must not be carried over.

Step 2. Mass ceiling and mapped-density bounds

Since $K_\sigma \leq 1$ pointwise and $P_\Gamma \leq P_{\max}$,

$$\phi^*(q, t) = \phi_0 + \int_{\Gamma(t)} K_\sigma(q - \xi) ds \leq \phi_0 + P_\Gamma \leq \phi_0 + P_{\max}$$

and $|V_i^\ell| \leq |B(p_i, R_\ell)| = \pi R_\ell^2$, hence

$$m_i^* \leq \pi R_\ell^2 (\phi_0 + P_{\max}) = m_{\max}$$

The measure decomposition of Lemma 1 bounds the mapped total weight by $P_{\max} + E_w$, which gives $\hat{\phi}_{\max}$ by the same argument. Differentiating under the integral sign and using $\|\nabla K_\sigma\|_\infty = e^{-1/2}/\sigma$ gives the Lipschitz constant $\hat{L}_\phi$.

Step 3. Midpoint mass error

Split the midpoint cells into those lying wholly inside V_i^ℓ and those cut by its boundary. For an interior cell Q with midpoint q_Q , Lipschitz continuity gives

$$\left| \int_Q \phi dq - \phi(q_Q) |Q| \right| \leq \hat{L}_\phi \max_{q \in Q} \|q - q_Q\| |Q| = \hat{L}_\phi r_h |Q|$$

because the farthest point of a cell from its midpoint is a corner, at distance $r_h = h/\sqrt{2}$. Summing over interior cells contributes at most $\hat{L}_\phi r_h A_V$.

The cut cells all lie within distance r_h of ∂V_i^ℓ . V_i^ℓ is convex, being an intersection of half-planes, a disk and a rectangle, so the inner and outer parallel sets of that width have combined area at most $2P_V r_h + \pi r_h^2$. Bounding the density there by $\hat{\phi}_{\max}$ and adding the two groups gives η_m . Convexity also justifies $P_V \leq 2\pi R_\ell$, the perimeter of the enclosing disk.

Step 4. First-moment error

Repeat Step 3 for the *shifted* first moment, which is what the centroid actually needs:

$$a = \int_{V_i^\ell} (q - p_i) \phi(q) dq$$

Its integrand $f(q) = (q - p_i)\phi(q)$ satisfies, on the cell,

$$\|f(q)\| \leq R_\ell \hat{\phi}_{\max}, \quad \text{Lip}(f) \leq \hat{\phi}_{\max} + R_\ell \hat{L}_\phi$$

since $\|q - p_i\| \leq R_\ell$ there. Substituting these two constants for $\hat{L}_\phi$ and $\hat{\phi}_{\max}$ in Step 3 yields $\|\hat{a} - a\| \leq \eta_a$.

Step 5. Ratio perturbation

The centroid is a ratio, so the two errors must be combined through a quotient bound rather than added. Write $\hat{c} - p_i = \hat{a}/\hat{m}$ and $c - p_i = a/m$. Then

$$\frac{\hat{a}}{\hat{m}} - \frac{a}{m} = \frac{\hat{a} - a}{\hat{m}} + a \left(\frac{1}{\hat{m}} - \frac{1}{m} \right) = \frac{\hat{a} - a}{\hat{m}} + \frac{a(m - \hat{m})}{m\hat{m}}$$

The gate $\eta_m < m_{\min}$ makes the denominator safely positive, $\hat{m} \geq m - \eta_m \geq m_{\min} - \eta_m > 0$, and $\|a\| \leq R_\ell m$ because $\|q - p_i\| \leq R_\ell$ on the cell. Therefore

$$\begin{aligned} \left\| \frac{\hat{a}}{\hat{m}} - \frac{a}{m} \right\| &\leq \frac{\|\hat{a} - a\|}{\hat{m}} + \frac{\|a\|}{m} \cdot \frac{|m - \hat{m}|}{\hat{m}} && \text{triangle inequality} \\ &\leq \frac{\eta_a}{m_{\min} - \eta_m} + R_\ell \frac{\eta_m}{m_{\min} - \eta_m} && \text{Steps 3-4 and } \|a\| \leq R_\ell m \\ &= \frac{\eta_a + R_\ell \eta_m}{m_{\min} - \eta_m} = \varepsilon_q && \text{definition} \end{aligned}$$

Step 6. Density perturbation on the same cell

Now hold the cell fixed and change only the density. Lemma 1 gives

$$|\tilde{m} - m^*| = \left| \int_{V_i^\ell} (\hat{\phi}_i - \phi^*) dq \right| \leq \varepsilon_\phi, \quad \|\tilde{a} - a^*\| = \left\| \int_{V_i^\ell} (q - p_i)(\hat{\phi}_i - \phi^*) dq \right\| \leq R_\ell \varepsilon_\phi$$

Both densities keep the base-density floor, so both masses are at least $m_{\min}$. Applying the same ratio identity as in Step 5,

$$\|\tilde{c} - c^*\| \leq \frac{R_\ell \varepsilon_\phi}{m_{\min}} + R_\ell \cdot \frac{\varepsilon_\phi}{m_{\min}} = \frac{2R_\ell \varepsilon_\phi}{m_{\min}}$$

Step 7. Assembly

Adding the density error of Step 6, the midpoint error of Step 5 and the separately certified geometric cell error gives (L2). ■

The denominator gate is fail-closed in the certificate generator: a configuration with $\eta_m \geq m_{\min}$ produces no certificate at all rather than a large one.

6 Proposition 1 - ideal Local-CVT gradient and descent

6.1 Statement

At every regular configuration covered by A1-A4 and A9,

$$\nabla_{p_i} H^*(P, t) = 2 m_i^* (p_i - c_i^*) \quad (\text{P1-a})$$

so the ideal Lloyd command is a scaled gradient step,

$$u_i^* = -k_c (p_i - c_i^*) = -\frac{k_c}{2m_i^*} \nabla_{p_i} H^*$$

and, with $A(P, t) = \text{diag}_i(\frac{k_c}{2m_i^*} I_2)$,

$$A(P, t) \succeq a_0 I, \quad a_0 = \frac{k_c}{2m_{\max}}, \quad \dot{H}^* = -\nabla H^{*\top} A \nabla H^* \leq -a_0 \|\nabla H^*\|^2 \quad (\text{P1-b})$$

the last identity holding with the reference frozen.

6.2 Proof

Step 1. Partition the integral

Away from the zero-measure tie and threshold sets of A9, the domain splits into the local cells and the saturated remainder $U = D \setminus \bigcup_j B(p_j, R_\ell)$, on which the running minimum equals R_ℓ^2 :

$$H^* = \sum_j \int_{V_j^\ell} \|q - p_j\|^2 \phi^* dq + R_\ell^2 \int_U \phi^* dq$$

Step 2. Volume term

Differentiating the integrand of the $j = i$ term with respect to p_i ,

$$\frac{\partial}{\partial p_i} \|q - p_i\|^2 = 2(p_i - q) \implies \int_{V_i^\ell} 2(p_i - q) \phi^* dq = 2 \left[m_i^* p_i - \int_{V_i^\ell} q \phi^* dq \right] = 2m_i^* (p_i - c_i^*)$$

No other volume term depends on p_i : for $j \neq i$ the integrand $\|q - p_j\|^2$ does not involve p_i , and the U integrand is the constant R_ℓ^2 .

Step 3. The moving-boundary terms cancel

This is the step that the saturated cost exists to make work. Moving p_i deforms ∂V_i^ℓ , so Reynolds' transport theorem contributes a flux for each moving piece of boundary. There are three kinds, and each vanishes for its own reason.

- *Shared Voronoi faces.* On the face between cells i and j we have $\|q - p_i\| = \|q - p_j\|$, so the two cells carry the same integrand value there. Whatever area cell i gains, cell j loses, with equal and opposite normal velocity; the two fluxes cancel exactly.
- *The moving circle* $\|q - p_i\| = R_\ell$. Just inside, the integrand is $\|q - p_i\|^2 = R_\ell^2$; just outside, the saturated value is also R_ℓ^2 . The integrand is continuous across the circle, so the flux cancels. Without saturation the outer value would be 0, the jump would be R_ℓ^2 , and a term proportional to $R_\ell^2 \oint \phi^*$ would survive.
- *The workspace boundary* ∂D is fixed, so its normal velocity is zero.

Only the volume derivative survives, which is (P1-a).

Step 4. Descent

By Lemma 2, $m_i^* \leq m_{\max}$, so every diagonal block satisfies $\frac{k_c}{2m_i^*} \geq \frac{k_c}{2m_{\max}} = a_0$, i.e. $A \succeq a_0 I$. Substituting $\dot{P} = -A \nabla H^*$ with the reference frozen,

$$\dot{H}^* = \nabla H^{*\top} \dot{P} = -\nabla H^{*\top} A \nabla H^* \leq -a_0 \|\nabla H^*\|^2 \leq 0$$

Step 5. The implemented cell is exact, not approximate

Suppose $q \in B(p_i, R_\ell)$ and agent j beats agent i at q , i.e. $\|q - p_j\| \leq \|q - p_i\| \leq R_\ell$. Then

$$\|p_j - p_i\| \leq \|p_j - q\| + \|q - p_i\| \leq 2R_\ell \leq R_{\text{comm}}$$

so every possible local competitor is already inside the communication neighbourhood. Under A2 the implemented cell therefore *equals* V_i^ℓ ; this is an identity, not an approximation. ■

IF $R_{\text{comm}} = R_\ell$

The argument above needs $2R_\ell$. A theorem-mode configuration with $R_{\text{comm}} < 2R_\ell$ is rejected by the validator, and the exact-cell claim would have to be marked BLOCKED rather than weakened silently.

Wolfram checked the integrand derivative and one exact fixed-square example. Those are sanity checks; the shape-derivative cancellation in Step 3 is the proof.

7 Proposition 2 - aggregate implementation disturbance

7.1 Statement

Write the realised stacked dynamics as an ideal field plus a disturbance:

$$\dot{P} = -A(P, t) \nabla H^*(P, t) + d(t), \quad \|d(t)\| \leq \delta$$

$$\delta = k_c \sqrt{\sum_{i=1}^N \varepsilon_{c,i}^2 + \varepsilon_{zoh} + \varepsilon_{num} + \varepsilon_{trk} + \varepsilon_{sf}} \quad (\text{P2})$$

7.2 Proof

Step 1. Centroid component

The nominal law and the ideal field differ only through the centroid each uses, so the gain multiplies the Lemma 2 error directly:

$$u_i^{\text{nom}} - u_i^* = -k_c(p_i - \hat{c}_i) + k_c(p_i - c_i^*) = k_c(\hat{c}_i - c_i^*) \implies \|u_i^{\text{nom}} - u_i^*\| \leq k_c \varepsilon_{c,i}$$

Stacking over the N agents in the frozen product norm,

$$\left\| (k_c(\hat{c}_i - c_i^*))_{i=1}^N \right\| = k_c \sqrt{\sum_i \|\hat{c}_i - c_i^*\|^2} \leq k_c \sqrt{\sum_i \varepsilon_{c,i}^2}$$

Step 2. Remaining deviations

Sample-and-hold, solver/integration, low-level tracking and safety-projection deviations are each bounded in the same stacked norm and added by the triangle inequality, which gives (P2).

Step 3. Analytic ceiling for the safety filter

The safety term admits a bound that needs no model of the filter at all, only the speed cap that both the nominal and the filtered command respect. If $\|u_i\| \leq u_{\max}$ for both, then

$$\|u_i^{\text{sf}} - u_i^{\text{nom}}\| \leq \|u_i^{\text{sf}}\| + \|u_i^{\text{nom}}\| \leq 2u_{\max} \implies \varepsilon_{\text{sf}} \leq \sqrt{N (2u_{\max})^2} = 2\sqrt{N} u_{\max}$$

The norm convention is read from the implementation, never guessed from the symbol. If the cap were componentwise, $\|u_i\|_{\infty} \leq u_{\max}$, then $\|u_i\|_2 \leq \sqrt{2}u_{\max}$ and the stacked ceiling would instead be $2\sqrt{2N} u_{\max}$. ■

THIS CEILING IS DELIBERATELY CRUDE

$2\sqrt{N}u_{\max}$ assumes the filter may reverse every command. A measured tightening may replace it only when its provenance and coverage interval are recorded in the ledger. Any theorem-mode gap, frontier or lead-side bias must be zero or appear as a further term in

(P2).

8 Proposition 3 - reference rate and local regularity

8.1 Target and density rate

Differentiating the offset target $\xi = b + d n$ by the product rule,

$$\dot{\xi} = \dot{b} + \dot{d} n + d \dot{n}$$

Under the rigid-body certificate A3, a point of the boundary at radius at most R_O from the object centre moves at speed at most $\bar{v}_O + \bar{\omega}_O R_O$, the unit normal rotates at rate at most $\bar{\omega}_O$, and the offset varies at rate at most $\bar{\omega}_O L_d$. Hence

$$\|\dot{\xi}\| \leq \bar{v}_O + \bar{\omega}_O (R_O + d_{\max} + L_d) =: v_{\xi, \max}$$

with $L_d = 0$ for the constant offset used in version 1. Differentiating the density under the integral sign and applying (T),

$$\partial_t \phi^*(q, t) = - \int_{\Gamma(t)} \nabla K_\sigma(q - \xi(s, t)) \cdot \dot{\xi}(s, t) ds \implies \nu_\phi := \|\partial_t \phi^*\|_1 \leq L_{K,1} P_{\max} v_{\xi, \max} \quad (\text{P3-a})$$

8.2 Cost and optimal-value rate

Because the saturated integrand is bounded, $0 \leq \min_i F_R \leq R_\ell^2$, the cost rate inherits the density rate:

$$|\partial_t H^*(P, t)| = \left| \int_D \min_i F_R(\|q - p_i\|) \partial_t \phi^*(q, t) dq \right| \leq R_\ell^2 \int_D |\partial_t \phi^*| dq = R_\ell^2 \nu_\phi$$

The optimal value inherits it too. For a fixed compact X and any two times, the elementary inequality $|\min f - \min g| \leq \sup |f - g|$ gives

$$|H_{\min}^*(t_2) - H_{\min}^*(t_1)| \leq \sup_{P \in X} |H^*(P, t_2) - H^*(P, t_1)| \leq R_\ell^2 \nu_\phi |t_2 - t_1|$$

so $H_{\min}^*$ is Lipschitz and, being Lipschitz, differentiable almost everywhere with $|\dot{H}_{\min}^*| \leq R_\ell^2 \nu_\phi$. Adding the two contributions bounds the drift of $V = H^* - H_{\min}^*$:

$$|\partial_t H^* - \dot{H}_{\min}^*| \leq \nu_H := 2R_\ell^2 \nu_\phi \quad (\text{P3-b})$$

RIGID BOUNDARIES ONLY

This derivation uses a rigid twist. Deformation, a time-varying perimeter, or split/merge events break it; those cases need a direct measure-rate assumption on $\partial_t \nu_t^*$ instead.

8.3 Local PL and quadratic growth (M5)

Suppose A8 holds: on a convex, symmetry-reduced chart of the selected branch, $\nabla^2 H_{\text{red}}^*(z, t) \succeq m_H I$ with $m_H > 0$. Write f for H^* in those coordinates and z^* for a minimiser. Integrating the Hessian twice along the segment from z^* to z gives the standard strong-convexity inequality

$$f(y) \geq f(z) + \nabla f(z)^\top (y - z) + \frac{m_H}{2} \|y - z\|^2 \quad \forall y, z$$

Putting $y = z^*$ gives quadratic growth,

$$H^* - H_{\min}^* \geq \frac{\alpha}{2} \text{dist}^2(P, \mathcal{C}^*(t)), \quad \alpha = m_H \quad (\text{QG})$$

and minimising the right-hand side over y instead, at $y = z - \nabla f(z)/m_H$, gives

$$\begin{aligned} H_{\min}^* &\geq H^*(z) - \frac{\|\nabla f(z)\|^2}{2m_H} && \text{minimise the quadratic lower bound over } y \\ \frac{1}{2} \|\nabla H^*\|^2 &\geq \mu [H^* - H_{\min}^*], \quad \mu = m_H && \text{rearrange} \end{aligned}$$

which is the local Polyak-Lojasiewicz inequality used in Theorem 1. ■

A8 IS CONDITIONAL, AND THAT IS THE HONEST STATUS

The implication above is proved. Its premise is not established here: no specific nondegenerate branch, cell combinatorics, reduced gauge, or complete parameter tube was supplied, so no validated interval enclosure of the reduced Hessian exists. A floating-point eigenvalue at one configuration would be NUMERIC-SANITY evidence and is not presented as proof. No global PL claim is made for a nonconvex CVT cost.

9 Theorem 1 - local practical stability

9.1 Statement

Assume A1-A10, Lemmas 1-2 and Propositions 1-3, on an interval throughout which the solution remains in the regular tube X . Define

$$V(P, t) = H^*(P, t) - H_{\min}^*(t) \geq 0, \quad a_0 = \frac{k_c}{2m_{\max}}, \quad \lambda_H = a_0\mu, \quad B = \frac{\delta^2}{2a_0} + \nu_H$$

Then for every $t \geq t_0$ in that interval

$$V(t) \leq e^{-\lambda_H(t-t_0)}V(t_0) + \frac{B}{\lambda_H}(1 - e^{-\lambda_H(t-t_0)}) \quad (\text{T1-a})$$

and, if the regular-tube hypotheses persist for all future time,

$$\limsup_{t \rightarrow \infty} \text{dist}(P(t), \mathcal{C}^*(t)) \leq \sqrt{\frac{2B}{\alpha \lambda_H}} \quad (\text{T1-b})$$

9.2 Proof - every inequality named

Differentiate V along the solution at an almost-every regular time and bound each term by the result that produces it.

$$\begin{aligned} \dot{V} &= \nabla H^{*\top} \dot{P} + \partial_t H^* - \dot{H}_{\min}^* && \text{chain rule; } H_{\min}^* \text{ differentiable a.e. by Prop. 3} \\ &= \nabla H^{*\top} (-A \nabla H^* + d) + \partial_t H^* - \dot{H}_{\min}^* && \text{substitute the dynamics} \\ &\leq -\nabla H^{*\top} A \nabla H^* + \|\nabla H^*\| \|d\| + \nu_H && \text{Cauchy-Schwarz, and (P3-b)} \\ &\leq -a_0 \|\nabla H^*\|^2 + \delta \|\nabla H^*\| + \nu_H && A \succeq a_0 I \text{ by (P1-b); } \|d\| \leq \delta \text{ by (P2)} \\ &\leq -a_0 \|\nabla H^*\|^2 + \frac{a_0}{2} \|\nabla H^*\|^2 + \frac{\delta^2}{2a_0} + \nu_H && \text{Young: } \delta x \leq \frac{a_0}{2} x^2 + \frac{\delta^2}{2a_0} \\ &= -\frac{a_0}{2} \|\nabla H^*\|^2 + B && \text{definition of } B \\ &\leq -a_0 \mu V + B = -\lambda_H V + B && \text{local PL: } \frac{1}{2} \|\nabla H^*\|^2 \geq \mu V \end{aligned}$$

The scalar comparison lemma applied to $\dot{V} \leq -\lambda_H V + B$ integrates this to (T1-a): the solution of the corresponding equality is exactly the right-hand side of (T1-a), and a differential inequality is dominated by it. Letting $t \rightarrow \infty$ in (T1-a), the exponential terms vanish and

$$\limsup_{t \rightarrow \infty} V(t) \leq \frac{B}{\lambda_H}$$

Finally, quadratic growth (QG) converts the Lyapunov level into a distance,

$$\text{dist}(P, \mathcal{C}^*) \leq \sqrt{\frac{2V}{\alpha}} \implies \limsup_{t \rightarrow \infty} \text{dist}(P(t), \mathcal{C}^*(t)) \leq \sqrt{\frac{2B}{\alpha \lambda_H}}$$

which is (T1-b). $\blacksquare$

9.3 Status

The implication is analytically complete and its scalar algebra is Wolfram-verified (Young, the comparison solution, and the strong convexity to PL step). The theorem remains **CONDITIONAL** on A5 and A8-A10 and on the run/tube certificates. It is local. It is not a global PL theorem, and it says nothing about caging or transport success.

10 Corollaries

10.1 Frozen perfect information

If every implementation error vanishes and the reference is frozen, then $\delta = 0$ and $\nu_H = 0$, so $B = 0$ and (T1-a) collapses to pure exponential decay:

$$\dot{V} \leq -\lambda_H V \implies V(t) \leq e^{-\lambda_H(t-t_0)} V(t_0) \implies \text{dist}(P(t), \mathcal{C}^*) \rightarrow 0$$

This is a sanity gate, not a decoration: the constant generator and the unit tests require exactly $B = 0$ and a zero ultimate radius in this case. If they did not, the formulas or their wiring would be wrong and Theorem 1 could not be marked complete.

10.2 Sampled-data corollary (M7)

Let the implementation take explicit steps of size Δt ,

$$P_{k+1} = P_k + \Delta t [-A_k g_k + d_k], \quad g_k = \nabla H^*(P_k, t_k), \quad a_0 I \preceq A_k \preceq a_1 I, \quad a_1 = \frac{k_c}{2m_{\min}}$$

with $\|d_k\| \leq \delta$ and ∇H^* spatially L_H -Lipschitz on X . Define

$$\bar{a} = a_0 - \frac{L_H a_1^2 \Delta t}{2}, \quad C_\delta = \delta(1 + L_H a_1 \Delta t), \quad B_d = \frac{C_\delta^2}{2\bar{a}} + \frac{L_H}{2} \delta^2 \Delta t + \nu_H$$

Then, under the step gates

$$\Delta t < \frac{2a_0}{L_H a_1^2}, \quad 0 < \bar{a} \mu \Delta t \leq 1 \quad (\text{M7-gate})$$

the Lyapunov level contracts geometrically,

$$V_{k+1} \leq (1 - \bar{a} \mu \Delta t) V_k + B_d \Delta t, \quad \limsup_{k \rightarrow \infty} \text{dist}(P_k, \mathcal{C}_k^*) \leq \sqrt{\frac{2B_d}{\alpha \bar{a} \mu}} \quad (\text{M7})$$

Proof

Apply the descent lemma to the L_H -Lipschitz gradient, add the reference drift over one step, and write $x = \|g_k\|$:

$$\begin{aligned}
V_{k+1} - V_k &\leq g_k^\top (P_{k+1} - P_k) + \frac{L_H}{2} \|P_{k+1} - P_k\|^2 + \nu_H \Delta t && \text{descent lemma and (P3-b)} \\
&\leq -a_0 \Delta t x^2 + \delta \Delta t x + \frac{L_H}{2} \Delta t^2 (a_1 x + \delta)^2 + \nu_H \Delta t && a_0 I \preceq A_k \preceq a_1 I, \|d_k\| \leq \delta \\
&= \Delta t \left[-\bar{a} x^2 + C_\delta x + \frac{L_H}{2} \delta^2 \Delta t + \nu_H \right] && \text{expand the square and collect powers of } x \\
&\leq \Delta t \left[-\frac{\bar{a}}{2} x^2 + \frac{C_\delta^2}{2\bar{a}} + \frac{L_H}{2} \delta^2 \Delta t + \nu_H \right] && \text{Young, with } \bar{a} > 0 \\
&= \Delta t \left[-\frac{\bar{a}}{2} x^2 + B_d \right] && \text{definition of } B_d \\
&\leq \Delta t [-\bar{a}\mu V_k + B_d] && \text{local PL: } \frac{1}{2}x^2 \geq \mu V_k
\end{aligned}$$

Rearranging gives the recursion in (M7). The first gate is exactly the condition $\bar{a} > 0$ needed for the Young step; the second keeps the contraction factor in $[0, 1)$. Summing the resulting geometric series,

$$\limsup_{k \rightarrow \infty} V_k \leq \frac{B_d \Delta t}{\bar{a}\mu \Delta t} = \frac{B_d}{\bar{a}\mu}$$

and quadratic growth converts this to the stated ultimate radius. ■

Wolfram expanded the descent algebra and checked the coefficient collection; a randomised positive-parameter test independently compares the certified bound against the direct one-step expression.

11 Limits and evidence status

Result	Status	Limiting condition
Lemma 1	PROVED + CAS	A5 map bounds and tail separation
Lemma 2	PROVED + CAS	rectangle, midpoint gate, exact-cell contract
Proposition 1	PROVED	regular tie/threshold sets; saturated cost
Proposition 2	PROVED	declared stacked disturbance components
Proposition 3	CONDITIONAL	rate proved; A8 branch coercivity assumed
Theorem 1	CONDITIONAL	complete implication under A1-A10
Sampled corollary	PROVED	explicit step gates

11.1 Evidence classes

- **PROVED**: the analytic implication is written out in this supplement.
- **VERIFIED**: Wolfram reproduced the identified integral or algebraic subclaim, and nothing beyond it.
- **CONDITIONAL**: a named branch, tube, map bound, or runtime deviation certificate is still required.
- **NUMERIC-SANITY**: debugging evidence only. No such result is used as proof anywhere in this package, and there are no such entries.

11.2 Fail-closed runtime gates

Each of these refuses to emit a certificate rather than emitting a degraded one, and each is covered by a test that feeds it a deliberately invalid configuration:

- **locality**: a theorem-mode configuration with $R_{\text{comm}} < 2R_\ell$ is rejected;
- **arc length**: a boundary observation without positive Δs_k is rejected, and the unit-weight fallback is forbidden;
- **quadrature rule**: anything other than a midpoint grid is rejected;
- **quadrature gate**: $\eta_m \geq m_{\text{min}}$ is rejected;
- **bias**: a nonzero gap or exploration gain is rejected unless separately budgeted into δ ;
- **sampled step**: Δt violating (M7-gate) is rejected.

11.3 Claim audit

This supplement contains no global PL statement, no formal caging claim, no finite-time phase-completion result, and no unconditional transport or safety guarantee. The numeric certificate in Appendix A validates the dependency wiring and the gates; its ultimate radius is deliberately conservative and may be numerically vacuous. It is not experimental evidence and is not used to claim transport performance.

A Constant ledger

Every value below is produced by one code path from the frozen formulas; none is entered by hand. The chain from raw sensing bounds to the ultimate radius is

$$(\varepsilon_b, \varepsilon_n, \varepsilon_d, v, h_s, E_w, M_{\text{drop}}) \longrightarrow \varepsilon_\phi \longrightarrow \varepsilon_c \longrightarrow \delta \longrightarrow B \longrightarrow \sqrt{\frac{2B}{\alpha\lambda_H}}$$

Constant	Value	Unit	Status
kernel_l1	0.251327	m ²	verified
kernel_grad_l1	1.57496	m	verified
kernel_grad_linf	3.03265	1/m	verified
epsilon_xi	0.00449999	m	conditional
epsilon_phi	0.106034	m ³	proved
m_min	5.02655	m ³	proved
m_max	34.5827	m ³	proved
grid_spacing	0.000390625	m	verified
eta_m	0.0598959	m ³	proved
eta_a	0.0574695	m ⁴	proved
epsilon_q	0.0212188	m	proved
epsilon_c	0.0549703	m	proved
delta	0.156932	m/s	conditional
nu_phi	0.464928	m ³ /s	proved
nu_H	0.595108	m ⁵ /s	proved
a0	0.0130123	1/(s*m ³)	proved
lambda_H	0.000650615	1/s	conditional
B	1.54143	m ⁵ /s	proved
ultimate_radius	307.843	m	conditional
sampled_Bd	1.54617	m ⁵ /s	proved
sampled_ultimate_radius	308.792	m	conditional

B Wolfram verification manifest

Connector: Wolfram (version not exposed). Local runtime: WolframScript 1.14.0; all nine committed scripts executed with exit code 0. Exact inputs, assumptions and outputs are archived under theory/wolfram/results.

ID	Target	Evidence	Verdict
W-L1-01/02	Gaussian L^1 integrals	EXACT	PASS
W-L1-03	gradient radial maximum	SYMBOLIC	PASS
W-L1-04/05	normal and bounded-step identities	SYMBOLIC	PASS
W-L2-01/02	ratio, monotonicity, mass bounds	SYMBOLIC	PASS
W-P1-01	integrand derivative, fixed-cell check	SYMBOLIC	PASS
W-P3-01	reference-rate algebra	SYMBOLIC	PASS
W-T1-01	Young and comparison ODE	SYMBOLIC	PASS
W-C1-01	sampled-data descent expansion	SYMBOLIC	PASS
W-M5-01	generic quadratic implication	SYMBOLIC	PASS
W-M5-02	actual branch Hessian box	INTERVAL	CONDITIONAL

Evidence rule: an exact or symbolic output supports only the stated integral or algebraic identity. Interval evidence would have to cover the complete selected tube to support A8, and none does.

B.1 What computer algebra did not verify

- the pushforward construction, the L^1 translation inequality, the curve quadrature and the atom matching of Lemma 1;
- the Voronoi shape derivative and the moving-boundary flux cancellation of Proposition 1;
- any uniform lower bound on a reduced Hessian for a named branch, which is why A8 and therefore Theorem 1 remain conditional.

B.2 Reference

J. Cortes, S. Martinez and F. Bullo, “Spatially-distributed coverage optimization and control with limited-range interactions”, ESAIM: COCV, 11(4):691-719, 2005. DOI 10.1051/cocv:2005024.